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# 1 Class Diagram

UML Type: Class Diagram Source: docs/diagrams/plant\_class\_diagram.puml

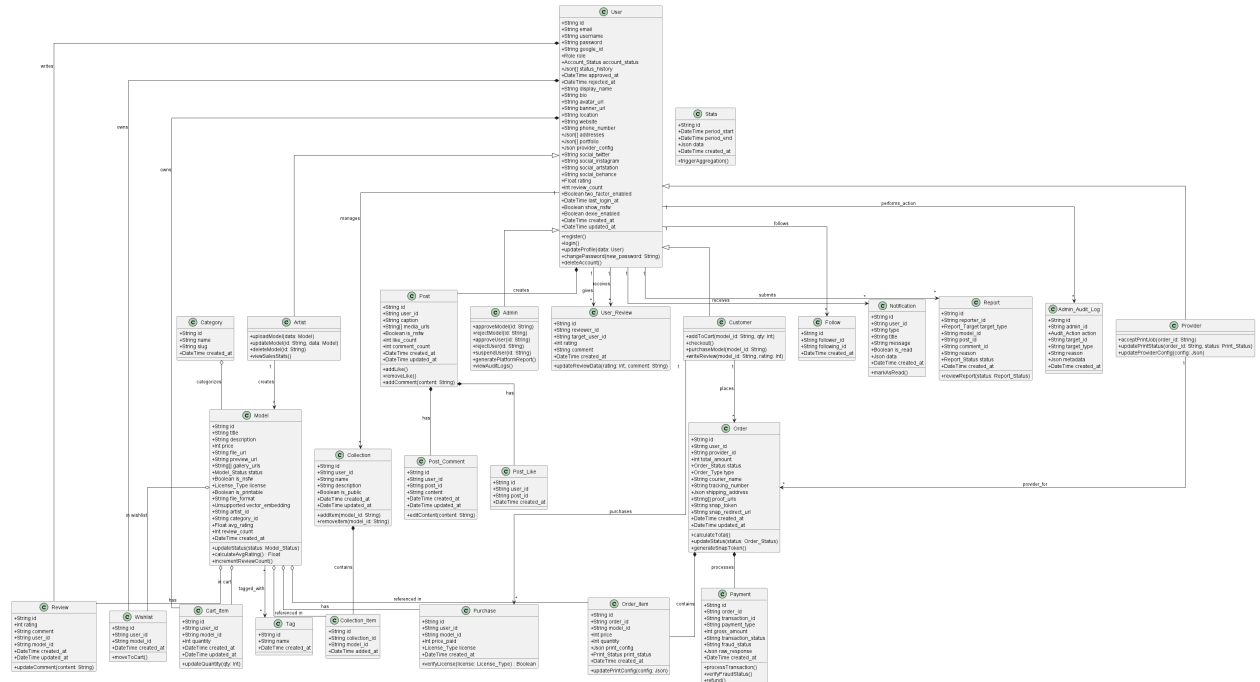


Figure 1: Class Diagram

## 1.1 Overview

This diagram describes the full object-oriented class model of the 3Dex platform. It defines each domain entity as a class with attributes and operations, then maps the inheritance hierarchy and associations among them.

## 1.2 Inheritance Hierarchy

All user roles extend a single base **User** class:

- **Customer** extends **User** - can shop, checkout, review, and write posts.
- **Artist** extends **User** - can upload and manage 3D models.
- **Provider** extends **User** - can manage print jobs.
- **Admin** extends **User** - can moderate content and users.

This matches the Prisma **Role** enum (**CUSTOMER**, **ARTIST**, **ADMIN**, **PROVIDER**) stored on the single **User** table.

## 1.3 Classes

### 1.3.1 User

The base entity for all platform actors.

| Field              | Type           | Description                                 |
|--------------------|----------------|---------------------------------------------|
| id                 | String         | UUID primary key                            |
| email              | String         | Unique login email                          |
| username           | String         | Unique handle                               |
| password           | String         | bcrypt-hashed; null for OAuth-only accounts |
| google_id          | String         | Unique Google OAuth subject ID              |
| role               | Role           | One of CUSTOMER, ARTIST, PROVIDER, ADMIN    |
| account_status     | Account_Status | PENDING, APPROVED, REJECTED                 |
| status_history     | Json[]         | Audit trail of status changes               |
| approved_at        | DateTime       | Timestamp of approval                       |
| rejected_at        | DateTime       | Timestamp of rejection                      |
| display_name       | String         | Public display name                         |
| bio                | String         | Profile bio                                 |
| avatar_url         | String         | S3 key for avatar image                     |
| banner_url         | String         | S3 key for profile banner                   |
| location           | String         | Free-text location                          |
| website            | String         | Personal website link                       |
| phone_number       | String         | Contact number                              |
| addresses          | Json[]         | Shipping addresses array                    |
| portfolio          | Json[]         | Portfolio items for Artists/Providers       |
| provider_config    | Json           | Print capabilities config for Providers     |
| social_twitter     | String         | Twitter/X handle                            |
| social_instagram   | String         | Instagram handle                            |
| social_artstation  | String         | ArtStation handle                           |
| social_behance     | String         | Behance handle                              |
| rating             | Float          | Aggregate rating from User_Review records   |
| review_count       | Int            | Total number of reviews received            |
| two_factor_enabled | Boolean        | 2FA preference flag                         |
| last_login_at      | DateTime       | Timestamp of most recent login              |
| show_nsfw          | Boolean        | NSFW content visibility preference          |
| dexie_enabled      | Boolean        | Whether Dexie AI assistant is active        |
| created_at         | DateTime       | Account creation timestamp                  |
| updated_at         | DateTime       | Last modification timestamp                 |

**Operations:** `register()`, `login()`, `updateProfile()`, `changePassword()`, `deleteAccount()`

### 1.3.2 Customer (extends User)

A base user who can browse, buy, and engage with content.

**Additional Operations:** `addToCart(model_id, qty)`, `checkout()`, `purchaseModel(model_id)`, `writeReview(model_id, rating)`

### 1.3.3 Artist (extends User)

A creator who uploads and sells 3D models. Requires admin approval (`account_status = APPROVED`).

**Additional Operations:** `uploadModel(data)`, `updateModel(id, data)`, `deleteModel(id)`, `viewSalesStats()`

### 1.3.4 Provider (extends User)

A 3D printing service provider. Also requires admin approval.

**Additional Operations:** `acceptPrintJob(order_id)`, `updatePrintStatus(order_id, status)`, `updateProviderConfig(config)`

---

### 1.3.5 Admin (extends User)

A platform moderator with elevated privileges.

**Additional Operations:** `approveModel(id)`, `rejectModel(id)`, `approveUser(id)`, `rejectUser(id)`, `suspendUser(id)`, `generatePlatformReport()`, `viewAuditLogs()`

---

### 1.3.6 Model

A 3D asset listed on the platform.

| Field        | Type         | Description                                 |
|--------------|--------------|---------------------------------------------|
| id           | String       | UUID primary key                            |
| title        | String       | Model name                                  |
| description  | String       | Optional description                        |
| price        | Int          | Price in IDR; 0 for free                    |
| file_url     | String       | S3 key for the GLB/GLTF file                |
| preview_url  | String       | S3 key for WebP preview image               |
| gallery_urls | String[]     | S3 keys for additional gallery images       |
| status       | Model_Status | PENDING, APPROVED, REJECTED                 |
| is_nsfw      | Boolean      | NSFW content flag                           |
| license      | License_Type | PERSONAL_USE or COMMERCIAL_USE              |
| is_printable | Boolean      | Whether the model supports print orders     |
| file_format  | String       | Default “glb”                               |
| embedding    | vector       | 384-dimension semantic embedding (pgvector) |
| artist_id    | String       | FK to User (Artist role)                    |
| category_id  | String       | FK to Category                              |
| avg_rating   | Float        | Aggregated average rating                   |
| review_count | Int          | Number of reviews                           |
| created_at   | DateTime     | Creation timestamp                          |

**Operations:** `updateStatus()`, `calculateAvgRating()`, `incrementReviewCount()`

---

### 1.3.7 Purchase

A permanent record of a user owning a model after payment.

| Field      | Type         | Description                               |
|------------|--------------|-------------------------------------------|
| id         | String       | UUID primary key                          |
| user_id    | String       | FK to User                                |
| model_id   | String       | FK to Model (nullable for deleted models) |
| price_paid | Int          | Price at time of purchase in IDR          |
| license    | License_Type | License type at time of purchase          |

| Field      | Type     | Description        |
|------------|----------|--------------------|
| created_at | DateTime | Purchase timestamp |

**Constraint:** Unique on (user\_id, model\_id).

### 1.3.8 Category

A taxonomy grouping for models.

| Field      | Type     | Notes               |
|------------|----------|---------------------|
| id         | String   | UUID primary key    |
| name       | String   | Unique display name |
| slug       | String   | Unique URL slug     |
| created_at | DateTime | Creation timestamp  |

### 1.3.9 Tag

A free-form label attached to models. Many-to-many with `Model`.

| Field      | Type     | Notes              |
|------------|----------|--------------------|
| id         | String   | UUID primary key   |
| name       | String   | Unique tag name    |
| created_at | DateTime | Creation timestamp |

### 1.3.10 Review

A model review written by a verified purchaser.

| Field      | Type     | Notes               |
|------------|----------|---------------------|
| id         | String   | UUID primary key    |
| rating     | Int      | 1-5 stars           |
| comment    | String   | Optional text       |
| user_id    | String   | FK to User          |
| model_id   | String   | FK to Model         |
| created_at | DateTime | Creation timestamp  |
| updated_at | DateTime | Last edit timestamp |

**Constraint:** Unique on (user\_id, model\_id).

### 1.3.11 Wishlist

A saved model reference per user. Toggle-based.

| Field      | Type     |
|------------|----------|
| id         | String   |
| user_id    | String   |
| model_id   | String   |
| created_at | DateTime |

**Constraint:** Unique on (user\_id, model\_id).

---

### 1.3.12 Cart\_Item

A staging area before checkout. All digital cart items have quantity 1.

| Field      | Type     |
|------------|----------|
| id         | String   |
| user_id    | String   |
| model_id   | String   |
| quantity   | Int      |
| created_at | DateTime |
| updated_at | DateTime |

**Constraint:** Unique on (user\_id, model\_id).

---

### 1.3.13 Order

Represents a checkout transaction, either for digital assets or print jobs.

| Field             | Type         | Description                                  |
|-------------------|--------------|----------------------------------------------|
| id                | String       | UUID primary key                             |
| user_id           | String       | FK to User (customer)                        |
| provider_id       | String       | FK to User (Provider); null for ASSET orders |
| total_amount      | Int          | Total in IDR                                 |
| status            | Order_Status | PENDING, PAID, FAILED, CANCELLED             |
| type              | Order_Type   | ASSET or PRINT_JOB                           |
| courier_name      | String       | Courier for print jobs                       |
| tracking_number   | String       | Tracking code for shipped print jobs         |
| shipping_address  | Json         | Address JSON for print delivery              |
| proof_urls        | String[]     | Photo evidence for shipped jobs              |
| snap_token        | String       | Midtrans Snap token                          |
| snap_redirect_url | String       | Midtrans redirect URL                        |
| created_at        | DateTime     | Creation timestamp                           |
| updated_at        | DateTime     | Last modification timestamp                  |

### 1.3.14 Order\_Item

A single line within an order referencing one model.

| Field        | Type         | Description                                                  |
|--------------|--------------|--------------------------------------------------------------|
| id           | String       | UUID primary key                                             |
| order_id     | String       | FK to Order                                                  |
| model_id     | String       | FK to Model (nullable if model deleted)                      |
| price        | Int          | Unit price in IDR at time of purchase                        |
| quantity     | Int          | Quantity (1 for digital, configurable for print)             |
| print_config | Json         | Material, infill, color, scale settings                      |
| print_status | Print_Status | PENDING, ACCEPTED, PROCESSING, SHIPPED, DELIVERED, CANCELLED |
| created_at   | DateTime     | Creation timestamp                                           |

### 1.3.15 Payment

A Midtrans transaction record linked to an order.

| Field              | Type     | Description                     |
|--------------------|----------|---------------------------------|
| id                 | String   | UUID primary key                |
| order_id           | String   | FK to Order                     |
| transaction_id     | String   | Unique Midtrans transaction ID  |
| payment_type       | String   | e.g. "credit_card", "gopay"     |
| gross_amount       | Int      | Total charged in IDR            |
| transaction_status | String   | Midtrans status string          |
| fraud_status       | String   | Midtrans fraud detection result |
| raw_response       | Json     | Full Midtrans webhook payload   |
| created_at         | DateTime | Webhook received timestamp      |

### 1.3.16 User\_Review

A review of an Artist or Provider by a verified transacting user.

| Field          | Type     | Notes                           |
|----------------|----------|---------------------------------|
| id             | String   | UUID primary key                |
| reviewer_id    | String   | FK to User (reviewer)           |
| target_user_id | String   | FK to User (artist or provider) |
| rating         | Int      | 1-5 stars                       |
| comment        | String   | Optional comment                |
| created_at     | DateTime | Creation timestamp              |

**Constraint:** Unique on (reviewer\_id, target\_user\_id).



### 1.3.17 Post

A community post created by Artists or Providers.

| Field         | Type     | Notes                       |
|---------------|----------|-----------------------------|
| id            | String   | UUID primary key            |
| user_id       | String   | FK to User                  |
| caption       | String   | Post text                   |
| media_urls    | String[] | S3 keys for uploaded images |
| is_nsfw       | Boolean  | NSFW flag                   |
| like_count    | Int      | Denormalized count          |
| comment_count | Int      | Denormalized count          |
| created_at    | DateTime | Creation timestamp          |
| updated_at    | DateTime | Last edit timestamp         |

### 1.3.18 Post\_Like

Join record between a User and a Post for the like feature.

| Field      | Type     |
|------------|----------|
| id         | String   |
| user_id    | String   |
| post_id    | String   |
| created_at | DateTime |

**Constraint:** Unique on (user\_id, post\_id).

### 1.3.19 Post\_Comment

A text comment attached to a Post.

| Field      | Type     |
|------------|----------|
| id         | String   |
| user_id    | String   |
| post_id    | String   |
| content    | String   |
| created_at | DateTime |
| updated_at | DateTime |

### 1.3.20 Stats

A periodic aggregation record created by the admin dashboard's aggregation trigger.

| Field        | Type     | Description              |
|--------------|----------|--------------------------|
| id           | String   | UUID primary key         |
| period_start | DateTime | Aggregation window start |

| Field      | Type     | Description                   |
|------------|----------|-------------------------------|
| period_end | DateTime | Aggregation window end        |
| data       | Json     | Aggregated metrics JSON       |
| created_at | DateTime | When this record was computed |

### 1.3.21 Follow

A directional social connection from one user to another.

| Field        | Type     |
|--------------|----------|
| id           | String   |
| follower_id  | String   |
| following_id | String   |
| created_at   | DateTime |

**Constraint:** Unique on (follower\_id, following\_id).

### 1.3.22 Notification

An in-app notification record delivered to a user.

| Field      | Type     | Description                  |
|------------|----------|------------------------------|
| id         | String   | UUID primary key             |
| user_id    | String   | FK to recipient User         |
| type       | String   | Notification category string |
| title      | String   | Short heading                |
| message    | String   | Body text                    |
| is_read    | Boolean  | Read state                   |
| data       | Json     | Optional metadata payload    |
| created_at | DateTime | Delivery timestamp           |

### 1.3.23 Collection

A user-curated named group of models.

| Field       | Type     |
|-------------|----------|
| id          | String   |
| user_id     | String   |
| name        | String   |
| description | String   |
| is_public   | Boolean  |
| created_at  | DateTime |
| updated_at  | DateTime |

### 1.3.24 Collection\_Item

A join record linking a Collection to a Model.

| Field         | Type     |
|---------------|----------|
| id            | String   |
| collection_id | String   |
| model_id      | String   |
| added_at      | DateTime |

**Constraint:** Unique on (collection\_id, model\_id).

### 1.3.25 Report

A user-submitted abuse report targeting a model, post, or comment.

| Field       | Type          | Description                     |
|-------------|---------------|---------------------------------|
| id          | String        | UUID primary key                |
| reporter_id | String        | FK to User who filed the report |
| target_type | Report_Target | MODEL, POST, or COMMENT         |
| model_id    | String        | FK to Model (optional)          |
| post_id     | String        | FK to Post (optional)           |
| comment_id  | String        | FK to Post_Comment (optional)   |
| reason      | String        | Free-text reason                |
| status      | Report_Status | PENDING, REVIEWED, DISMISSED    |
| created_at  | DateTime      | Submission timestamp            |

### 1.3.26 Admin\_Audit\_Log

An immutable record of every admin action taken on the platform.

| Field       | Type         | Description                                                                                                                       |
|-------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| id          | String       | UUID primary key                                                                                                                  |
| admin_id    | String       | FK to User (Admin role)                                                                                                           |
| action      | Audit_Action | DELETE_MODEL,<br>DELETE_POST,<br>DELETE_COMMENT,<br>BAN_USER,<br>REJECT_MODEL,<br>REJECT_USER,<br>APPROVE_USER,<br>USER_STEP_DOWN |
| target_id   | String       | ID of the affected entity                                                                                                         |
| target_type | String       | Human-readable entity type<br>string                                                                                              |
| reason      | String       | Mandatory justification                                                                                                           |
| metadata    | Json         | Extra context (title, username,<br>price, etc.)                                                                                   |

| Field      | Type     | Description      |
|------------|----------|------------------|
| created_at | DateTime | Action timestamp |

## 1.4 Associations Summary

| From       | Relationship           | To                                                                 | Notes                                               |
|------------|------------------------|--------------------------------------------------------------------|-----------------------------------------------------|
| User       | owns (composition)     | Cart_Item, Wishlist, Post                                          | Deleted when user deleted                           |
| Order      | contains (composition) | Order_Item, Payment                                                | Part of order                                       |
| Post       | has (composition)      | Post_Like, Post_Comment                                            | Part of post                                        |
| Collection | contains (composition) | Collection_Item                                                    | Part of collection                                  |
| Artist     | creates                | Model                                                              | One Artist, many Models                             |
| Category   | categorizes            | Model                                                              | Optional FK                                         |
| Model      | referenced in          | Order_Item, Collection_Item, Review, Purchase, Wishlist, Cart_Item | Nullable after deletion                             |
| Customer   | purchases              | Purchase                                                           |                                                     |
| Customer   | places                 | Order                                                              |                                                     |
| Provider   | fulfills               | Order                                                              | provider_id FK                                      |
| User       | gives / receives       | User_Review                                                        | Self-referential via reviewer_id and target_user_id |
| User       | follows                | Follow                                                             | Directional social graph                            |
| User       | receives               | Notification                                                       |                                                     |
| User       | manages                | Collection                                                         |                                                     |
| User       | submits                | Report                                                             |                                                     |
| Admin      | performs_action        | Admin_Audit_Log                                                    |                                                     |
| Model      | tagged_with            | Tag                                                                | Many-to-many via implicit join                      |

## 2 Entity Relationship Diagram

**UML Type:** Entity Relationship Diagram (ERD) **Source:** docs/diagrams/plant\_erd.puml

### 2.1 Overview

This ERD maps the full relational structure of the 3Dex PostgreSQL database as modeled via Prisma. Every entity corresponds directly to a Prisma model in `schema.prisma`. Primary keys are UUIDs. Required fields are marked with an asterisk in the source file.



| Column        | Type     | Constraint | Notes                     |
|---------------|----------|------------|---------------------------|
| show_nsfw     | Boolean  |            | Content filter preference |
| dexie_enabled | Boolean  |            | Dexie AI toggle           |
| created_at    | DateTime | NOT NULL   |                           |
| updated_at    | DateTime | NOT NULL   | Auto-managed              |

### 2.2.2 Model

A 3D asset uploaded by an Artist.

| Column       | Type         | Constraint     | Notes                                |
|--------------|--------------|----------------|--------------------------------------|
| id           | String       | PK             | UUID                                 |
| title        | String       | NOT NULL       |                                      |
| description  | String       |                |                                      |
| price        | Int          | NOT NULL       | IDR; 0 = free                        |
| file_url     | String       | NOT NULL       | S3 key for GLB/GLTF                  |
| preview_url  | String       |                | S3 key for WebP preview              |
| gallery_urls | String[]     |                | Additional image keys                |
| status       | Model_Status |                | PENDING, APPROVED, REJECTED          |
| is_nsfw      | Boolean      |                |                                      |
| license      | License_Type |                | PERSONAL_USE, COMMERCIAL_USE         |
| is_printable | Boolean      |                |                                      |
| file_format  | String       |                | Default “glb”                        |
| embedding    | vector       |                | 384-dim pgvector for semantic search |
| artist_id    | String       | FK -> User     | Required                             |
| category_id  | String       | FK -> Category | Optional                             |
| avg_rating   | Float        |                | Aggregated rating                    |
| review_count | Int          |                | Denormalized count                   |
| created_at   | DateTime     | NOT NULL       |                                      |

### 2.2.3 Purchase

Immutable record created when payment is confirmed for a digital model.

| Column     | Type         | Constraint             |
|------------|--------------|------------------------|
| id         | String       | PK                     |
| user_id    | String       | FK -> User             |
| model_id   | String       | FK -> Model (nullable) |
| price_paid | Int          | NOT NULL               |
| license    | License_Type | NOT NULL               |
| created_at | DateTime     | NOT NULL               |

**Unique:** (user\_id, model\_id)

### 2.2.4 Category

A top-level taxonomy for models.

| Column     | Type     | Constraint |
|------------|----------|------------|
| id         | String   | PK         |
| name       | String   | UK         |
| slug       | String   | UK         |
| created_at | DateTime | NOT NULL   |

---

### 2.2.5 Tag

A label for filtering and semantic grouping.

| Column     | Type     | Constraint |
|------------|----------|------------|
| id         | String   | PK         |
| name       | String   | UK         |
| created_at | DateTime | NOT NULL   |

---

### 2.2.6 Review

A star rating and optional comment left by a purchaser.

| Column     | Type     | Constraint  |
|------------|----------|-------------|
| id         | String   | PK          |
| rating     | Int      | NOT NULL    |
| comment    | String   |             |
| user_id    | String   | FK -> User  |
| model_id   | String   | FK -> Model |
| created_at | DateTime | NOT NULL    |
| updated_at | DateTime | NOT NULL    |

**Unique:** (user\_id, model\_id)

---

### 2.2.7 Wishlist

A saved reference to a model.

| Column     | Type     | Constraint  |
|------------|----------|-------------|
| id         | String   | PK          |
| user_id    | String   | FK -> User  |
| model_id   | String   | FK -> Model |
| created_at | DateTime | NOT NULL    |

**Unique:** (user\_id, model\_id)

---

### 2.2.8 Cart\_Item

A model staged for checkout.

| Column     | Type     | Constraint  |
|------------|----------|-------------|
| id         | String   | PK          |
| user_id    | String   | FK -> User  |
| model_id   | String   | FK -> Model |
| quantity   | Int      | NOT NULL    |
| created_at | DateTime | NOT NULL    |
| updated_at | DateTime | NOT NULL    |

**Unique:** (user\_id, model\_id)

### 2.2.9 Order

An order for digital assets or a 3D print job.

| Column            | Type         | Constraint | Notes                            |
|-------------------|--------------|------------|----------------------------------|
| id                | String       | PK         | UUID                             |
| user_id           | String       | FK -> User | Buyer                            |
| provider_id       | String       | FK -> User | Provider; null for ASSET orders  |
| total_amount      | Int          | NOT NULL   | IDR                              |
| status            | Order_Status | NOT NULL   | PENDING, PAID, FAILED, CANCELLED |
| type              | Order_Type   | NOT NULL   | ASSET or PRINT_JOB               |
| courier_name      | String       |            | Print orders only                |
| tracking_number   | String       |            | Print orders only                |
| shipping_address  | Json         |            | Print orders only                |
| proof_urls        | String[]     |            | Photo proofs for shipped jobs    |
| snap_token        | String       |            | Midtrans token                   |
| snap_redirect_url | String       |            | Midtrans redirect                |
| created_at        | DateTime     | NOT NULL   |                                  |
| updated_at        | DateTime     | NOT NULL   |                                  |

### 2.2.10 Order\_Item

A single line item within an order.

| Column       | Type         | Constraint  | Notes                            |
|--------------|--------------|-------------|----------------------------------|
| id           | String       | PK          | UUID                             |
| order_id     | String       | FK -> Order |                                  |
| model_id     | String       | FK -> Model | Nullable if model deleted        |
| price        | Int          | NOT NULL    | IDR at time of purchase          |
| quantity     | Int          | NOT NULL    |                                  |
| print_config | Json         |             | Material, color, infill settings |
| print_status | Print_Status | NOT NULL    | PENDING through DELIVERED        |
| created_at   | DateTime     | NOT NULL    |                                  |



### 2.2.11 Payment

A Midtrans transaction record.

| Column             | Type     | Constraint  |
|--------------------|----------|-------------|
| id                 | String   | PK          |
| order_id           | String   | FK -> Order |
| transaction_id     | String   | UK          |
| payment_type       | String   | NOT NULL    |
| gross_amount       | Int      | NOT NULL    |
| transaction_status | String   | NOT NULL    |
| fraud_status       | String   |             |
| raw_response       | Json     |             |
| created_at         | DateTime | NOT NULL    |

### 2.2.12 User\_Review

A rating of an Artist or Provider by a verified customer.

| Column         | Type     | Constraint |
|----------------|----------|------------|
| id             | String   | PK         |
| reviewer_id    | String   | FK -> User |
| target_user_id | String   | FK -> User |
| rating         | Int      | NOT NULL   |
| comment        | String   |            |
| created_at     | DateTime | NOT NULL   |

**Unique:** (reviewer\_id, target\_user\_id)

### 2.2.13 Post

A community post with images and an optional caption.

| Column        | Type     | Constraint |
|---------------|----------|------------|
| id            | String   | PK         |
| user_id       | String   | FK -> User |
| caption       | String   |            |
| media_urls    | String[] |            |
| is_nsfw       | Boolean  |            |
| like_count    | Int      | NOT NULL   |
| comment_count | Int      | NOT NULL   |
| created_at    | DateTime | NOT NULL   |
| updated_at    | DateTime | NOT NULL   |

### 2.2.14 Post\_Like

Join table for the like relationship.

| Column     | Type     | Constraint |
|------------|----------|------------|
| id         | String   | PK         |
| user_id    | String   | FK -> User |
| post_id    | String   | FK -> Post |
| created_at | DateTime | NOT NULL   |

**Unique:** (user\_id, post\_id)

---

### 2.2.15 Post\_Comment

A text comment on a post.

| Column     | Type     | Constraint |
|------------|----------|------------|
| id         | String   | PK         |
| user_id    | String   | FK -> User |
| post_id    | String   | FK -> Post |
| content    | String   | NOT NULL   |
| created_at | DateTime | NOT NULL   |
| updated_at | DateTime | NOT NULL   |

### 2.2.16 Stats

A point-in-time aggregation snapshot.

| Column       | Type     | Constraint |
|--------------|----------|------------|
| id           | String   | PK         |
| period_start | DateTime | NOT NULL   |
| period_end   | DateTime | NOT NULL   |
| data         | Json     | NOT NULL   |
| created_at   | DateTime | NOT NULL   |

### 2.2.17 Follow

A directed edge in the social graph.

| Column       | Type     | Constraint |
|--------------|----------|------------|
| id           | String   | PK         |
| follower_id  | String   | FK -> User |
| following_id | String   | FK -> User |
| created_at   | DateTime | NOT NULL   |

**Unique:** (follower\_id, following\_id)

---

### 2.2.18 Notification

An in-app notification pushed to a user.

| Column     | Type     | Constraint |
|------------|----------|------------|
| id         | String   | PK         |
| user_id    | String   | FK -> User |
| type       | String   | NOT NULL   |
| title      | String   | NOT NULL   |
| message    | String   | NOT NULL   |
| is_read    | Boolean  | NOT NULL   |
| data       | Json     |            |
| created_at | DateTime | NOT NULL   |

### 2.2.19 Collection

A named list of models curated by a user.

| Column      | Type     | Constraint |
|-------------|----------|------------|
| id          | String   | PK         |
| user_id     | String   | FK -> User |
| name        | String   | NOT NULL   |
| description | String   |            |
| is_public   | Boolean  | NOT NULL   |
| created_at  | DateTime | NOT NULL   |
| updated_at  | DateTime | NOT NULL   |

### 2.2.20 Collection\_Item

Many-to-many join between Collection and Model.

| Column        | Type     | Constraint       |
|---------------|----------|------------------|
| id            | String   | PK               |
| collection_id | String   | FK -> Collection |
| model_id      | String   | FK -> Model      |
| added_at      | DateTime | NOT NULL         |

**Unique:** (collection\_id, model\_id)

### 2.2.21 Report

An abuse report submitted by a user.

| Column | Type   | Constraint |
|--------|--------|------------|
| id     | String | PK         |

| Column      | Type          | Constraint                    |
|-------------|---------------|-------------------------------|
| reporter_id | String        | FK -> User                    |
| target_type | Report_Target | NOT NULL                      |
| model_id    | String        | FK -> Model (optional)        |
| post_id     | String        | FK -> Post (optional)         |
| comment_id  | String        | FK -> Post_Comment (optional) |
| reason      | String        | NOT NULL                      |
| status      | Report_Status | NOT NULL                      |
| created_at  | DateTime      | NOT NULL                      |

### 2.2.22 Admin\_Audit\_Log

An append-only record of every admin action.

| Column      | Type         | Constraint |
|-------------|--------------|------------|
| id          | String       | PK         |
| admin_id    | String       | FK -> User |
| action      | Audit_Action | NOT NULL   |
| target_id   | String       |            |
| target_type | String       |            |
| reason      | String       |            |
| metadata    | Json         |            |
| created_at  | DateTime     | NOT NULL   |

## 2.3 Relationship Summary

| Table A | Cardinality  | Table B         | Via                            |
|---------|--------------|-----------------|--------------------------------|
| User    | 1 to many    | Model           | artist_id                      |
| User    | 1 to many    | Purchase        | user_id                        |
| User    | 1 to many    | Review          | user_id                        |
| User    | 1 to many    | Wishlist        | user_id                        |
| User    | 1 to many    | Cart_Item       | user_id                        |
| User    | 1 to many    | Order           | user_id (customer)             |
| User    | 1 to many    | Order           | provider_id (provider)         |
| User    | many to many | User_Review     | reviewer_id and target_user_id |
| User    | 1 to many    | Post            | user_id                        |
| User    | 1 to many    | Post_Like       | user_id                        |
| User    | 1 to many    | Post_Comment    | user_id                        |
| User    | many to many | Follow          | follower_id and following_id   |
| User    | 1 to many    | Notification    | user_id                        |
| User    | 1 to many    | Collection      | user_id                        |
| User    | 1 to many    | Report          | reporter_id                    |
| User    | 1 to many    | Admin_Audit_Log | admin_id                       |
| Model   | many to 1    | Category        | category_id                    |
| Model   | many to many | Tag             | implicit join table            |
| Model   | 1 to many    | Purchase        | model_id                       |
| Model   | 1 to many    | Review          | model_id                       |

| Table A    | Cardinality | Table B         | Via           |
|------------|-------------|-----------------|---------------|
| Model      | 1 to many   | Wishlist        | model_id      |
| Model      | 1 to many   | Cart_Item       | model_id      |
| Model      | 1 to many   | Order_Item      | model_id      |
| Model      | 1 to many   | Collection_Item | model_id      |
| Order      | 1 to many   | Order_Item      | order_id      |
| Order      | 1 to many   | Payment         | order_id      |
| Post       | 1 to many   | Post_Like       | post_id       |
| Post       | 1 to many   | Post_Comment    | post_id       |
| Collection | 1 to many   | Collection_Item | collection_id |

### 3 Use Case Diagram

UML Type: Use Case Diagram Source: docs/diagrams/3dex-ucd.drawio.xml

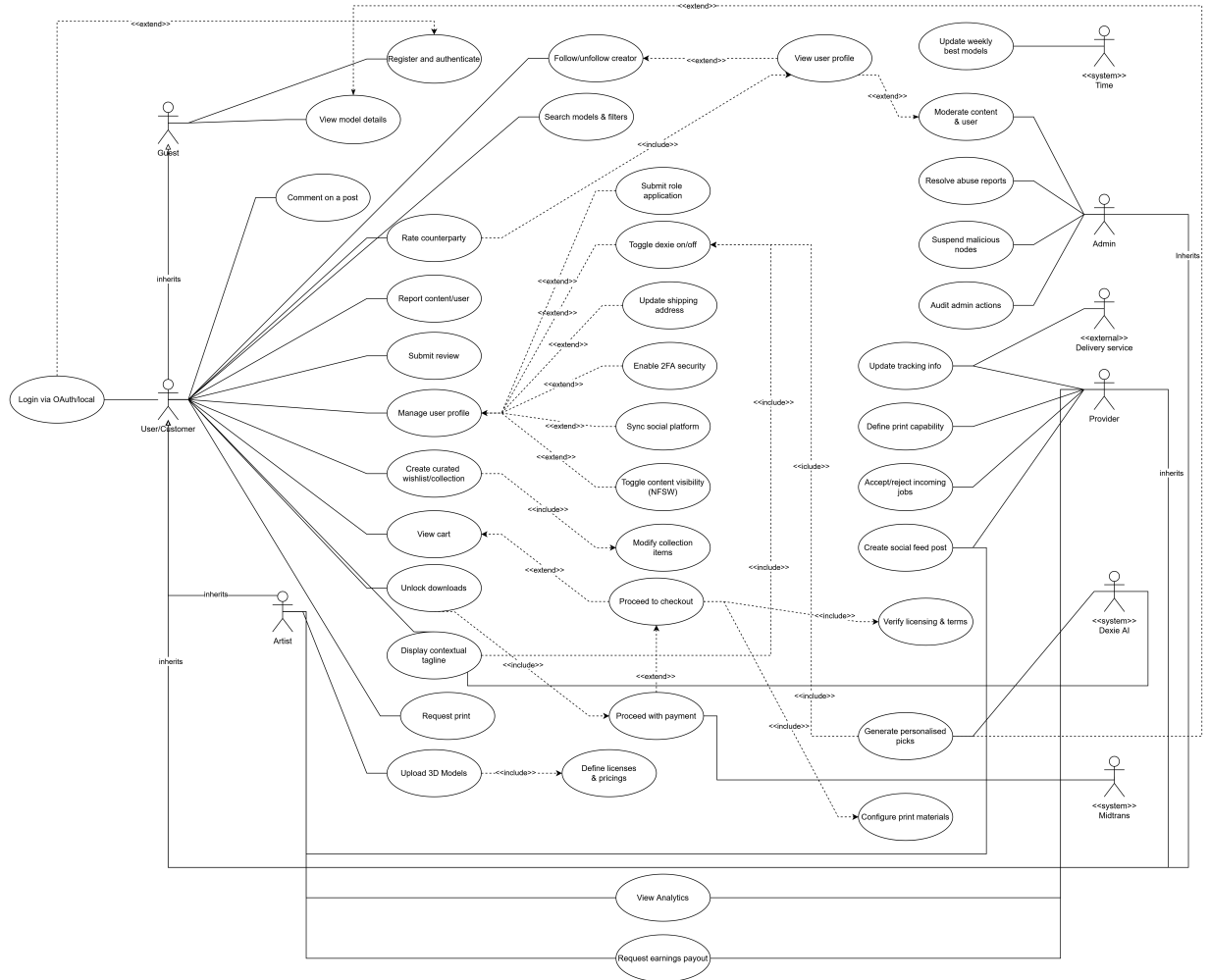


Figure 3: Use Case Diagram

## 3.1 Overview

This use case diagram represents the comprehensive behavioral model of the 3Dex platform. It maps the complete range of system capabilities to human and external system actors. The diagram separates functions into logical sub-domains and establishes precise <<include>> and <<extend>> relationships to represent execution logic.

---

## 3.2 Actors

The 3Dex platform categorizes actors into human participants (inheriting base capabilities in a structured hierarchy) and external system components.

### 3.2.1 Human Actors

| Actor                | Description                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Guest</b>         | An unauthenticated, anonymous visitor. Can perform base browsing activities like searching and viewing.                                                                |
| <b>User/Customer</b> | A registered, authenticated user. Serves as the base role for all active human participants. Inherits all capabilities of <b>Guest</b> .                               |
| <b>Artist</b>        | A registered creator authorized to upload, manage, and sell 3D models. Inherits all capabilities of <b>User/Customer</b> .                                             |
| <b>Provider</b>      | A registered 3D printing service provider who handles local fabrication and delivery of print orders. Inherits all capabilities of <b>User/Customer</b> .              |
| <b>Admin</b>         | The system administrator who moderates content, processes applications, manages system health, and audits actions. Inherits all capabilities of <b>User/Customer</b> . |

### 3.2.2 External & System Actors

| Actor                     | Type                     | Description                                                                                      |
|---------------------------|--------------------------|--------------------------------------------------------------------------------------------------|
| > <b>Midtrans</b>         | External Payment Gateway | Responsible for secure transaction processing and payment status webhooks.                       |
| > <b>Dexie AI</b>         | Gemini-powered Subsystem | Handles intelligent tagline generation and personalized recommendation queries.                  |
| > <b>Delivery service</b> | Logistics Gateway        | Receives and updates tracking information for physical 3D print products.                        |
| > <b>Time</b>             | Chron Scheduler          | Automated timer that triggers recurring background operations (e.g., weekly best models update). |

### 3.3 Use Cases by Domain

#### 3.3.1 Identity & Profile Management

- **Register and authenticate:** Accessible by **Guest** to establish a new account.
- **Login via OAuth/local:** Included in **Register and authenticate** to authenticate users via local credentials or Google OAuth.
- **Manage user profile:** Central use case for modifying personal information.
  - *Extensions:*
    - \* **Update shipping address:** Adds physical shipping details to the user profile.
    - \* **Enable 2FA security:** Activates two-factor authentication (2FA).
    - \* **Sync social platform:** Connects portfolios (Behance, Artstation, Twitter, etc.).
    - \* **Toggle content visibility (NFSW):** Sets content filters.
    - \* **Toggle dexie on/off:** Persists AI preferences.
    - \* **Submit role application:** Allows standard customers to apply to become an Artist or Provider.

#### 3.3.2 Catalog & Discovery

- **Search models & filters:** Allows searching via categories, pricing, and tag filters.
- **View model details:** High-fidelity Three.js-based viewer page.
- **Display contextual tagline:** Dynamically generates routes/headers via > **Dexie AI**. Extends **Toggle dexie on/off**.
- **Generate personalised picks:** Taste-based recommendations using pgvector similarity. Uses > **Dexie AI** and extends **Toggle dexie on/off** and **View model details**.
- **View best model of the week:** Highlighted showcase on the home page.

#### 3.3.3 Social & Feedback

- **Follow/unfollow creator:** Subscribes to creators to receive feed updates.
- **Create social feed post:** Enabled for **Artist** and **Provider** to publish updates.
- **Like/Comment:** Allows liking and commenting on community posts.
- **Rate counterparty:** Submits provider or artist reviews after a completed print job or model purchase.
- **Submit review:** Submits detailed reviews for models.
- **Report content/user:** Submits abuse or moderation reports.

#### 3.3.4 Curation & Shopping

- **Create curated wishlist/collection:** Allows users to save models.
  - *Includes:*
    - \* **Modify collection items:** Adding or removing models from specific folders.
- **View cart:** Interactive checkout preparation stage.
- **Proceed to checkout:** Initiates checkout processing.
  - *Includes:*
    - \* **Verify licensing & terms:** Ensures valid license type selection (Personal vs. Commercial).
  - *Extensions:*
    - \* **Configure print materials:** Configures infill, colors, and scaling for physical 3D print requests.

#### 3.3.5 Financial & Fulfillment

- **Proceed with payment:** Handles order creation and token acquisition via > **Midtrans**.
- **Unlock downloads:** Grants pre-signed S3 download URLs upon verified order completion.
- **Request earnings payout:** Enables **Artist** and **Provider** to withdraw accumulated earnings.
- **Define print capability:** Enabled for **Provider** to manage capabilities (materials, colors, local pricing).

- **Accept/reject incoming jobs:** Managed by **Provider** to claim print jobs.
- **Update tracking info:** Enables **Provider** and **> Delivery service** to record shipping references.
- **View Delivery Status:** Allows users to track physical shipping progress.
- **View Analytics:** Allows **Artist** and **Provider** to track monthly revenues and performance.

### 3.3.6 Administrative Operations

- **Moderate content & user:** General content and account lifecycle moderation.
- **Resolve abuse reports:** Reviews and closes submitted reports.
- **Suspend malicious nodes:** Suspends offending users or bans specific items.
- **Audit admin actions:** Records all administrative changes to an append-only ledger (**Admin\_Audit\_Log**).
- **Manage platform stats:** Views system revenue and model performance snapshots.

## 3.4 Core Relationships

### 3.4.1 Include Relationships (<<include>>)

| Base Use Case                             | Included Use Case                     | Rationale                                                                           |
|-------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------|
| <b>Register and authenticate</b>          | <b>Login via OAuth/local</b>          | Every new registration session concludes with active authentication.                |
| <b>Create curated wishlist/collection</b> | <b>Modify collection items</b>        | Saving a collection implies the mandatory capacity to structure and edit its items. |
| <b>Upload 3D Models</b>                   | <b>Define licenses &amp; pricings</b> | All catalog uploads must have designated licensing terms and prices.                |
| <b>Proceed to checkout</b>                | <b>Verify licensing &amp; terms</b>   | Checking out requires explicit acceptance of the specific license.                  |

### 3.4.2 Extend Relationships (<<extend>>)

| Extension Use Case                      | Base Use Case              | Extension Point / Condition                                            |
|-----------------------------------------|----------------------------|------------------------------------------------------------------------|
| <b>Update shipping address</b>          | <b>Manage user profile</b> | Triggered conditionally when shipping fields are populated.            |
| <b>Enable 2FA security</b>              | <b>Manage user profile</b> | Triggered conditionally when user toggles 2FA security.                |
| <b>Sync social platform</b>             | <b>Manage user profile</b> | Triggered conditionally when linking external accounts.                |
| <b>Toggle content visibility (NFSW)</b> | <b>Manage user profile</b> | Triggered conditionally when editing sensitive filters.                |
| <b>Toggle dexie on/off</b>              | <b>Manage user profile</b> | Triggered conditionally when customising AI preferences.               |
| <b>Submit role application</b>          | <b>Manage user profile</b> | Triggered when a customer applies for Artist/Provider status.          |
| <b>Configure print materials</b>        | <b>Proceed to checkout</b> | Triggered when the checkout payload contains a physical print job.     |
| <b>Display contextual tagline</b>       | <b>Toggle dexie on/off</b> | Active only when the Dexie AI system toggle is set to <b>enabled</b> . |



| Extension Use Case                 | Base Use Case              | Extension Point / Condition                                            |
|------------------------------------|----------------------------|------------------------------------------------------------------------|
| <b>Generate personalised picks</b> | <b>Toggle dexie on/off</b> | Active only when the Dexie AI system toggle is set to <b>enabled</b> . |
| <b>Generate personalised picks</b> | <b>View model details</b>  | Triggered conditionally when displaying similarity recommendations.    |

### 3.4.3 External System Integrations

- **Midtrans Gateway:**
  - Invoked by **Proceed with payment** to handle transaction processing.
  - Triggers **Unlock downloads** upon receiving a verified payment webhook.
- **Dexie AI Gateway:**
  - Triggers **Display contextual tagline** and **Generate personalised picks** to inject Gemini-powered recommendations into active browsing contexts.
- **Delivery Service:**
  - Triggers **Update tracking info** to sync physical courier status back to the order record.
- **Time Scheduler:**
  - Triggers **Update weekly best models** dynamically every calendar interval.